

THE EXACT SCHATTEN $3/2$ CONSTANT FOR TWO RANK-ONE MATRICES

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ABSTRACT. We determine the exact best constant in

$$\|A + B\|_{3/2} \leq C \| |A| + |B| \|_{3/2}$$

when A and B have rank at most one. For complex matrices of every size $n \geq 2$, the sharp dimension-free value is

$$C_* = \left(\frac{107 + 51\sqrt{17}}{256} \right)^{1/6} = 1.0364143573334075 \dots$$

It is already attained by real 2×2 matrices. We classify all nonzero equality cases up to the natural two-sided unitary, scaling, phase, and permutation symmetries. The two decisive correlations are algebraic: their squares are $(\sqrt{17} - 1)/8$ and $(5\sqrt{17} - 13)/8$. The apparent three-parameter optimization collapses, through Schatten duality, weighted Cauchy–Schwarz, and Hölder’s inequality, to a one-variable maximization over the eigenvalue ratio of $|A| + |B|$. Finally, we prove by exact rational inequalities that C_* is strictly larger than the value proposed by the Tang–Zhang formula at $p = 3/2, m = 2$, and provide a standard-library exact certificate.

1. INTRODUCTION

For $X \in \mathbb{M}_n(\mathbb{C})$, write

$$|X| = (X^* X)^{1/2}, \quad \|X\|_p = (\operatorname{Tr} |X|^p)^{1/p} \quad (1 \leq p < \infty).$$

Tang and Zhang [2] asked for the best constants in

$$\left\| \sum_{k=1}^m A_k \right\|_p \leq c_p(m) \left\| \sum_{k=1}^m |A_k| \right\|_p \quad (1)$$

and conjectured a closed formula for finite $p > 1$. If $x_{p,m} > 1$ denotes the unique root of

$$x^p - 2x - (m - 1) = 0,$$

their proposed value is

$$C_{p,m}^{\text{TZ}} = \frac{\sqrt{x_{p,m}(x_{p,m} + m - 1)}}{(x_{p,m}^p + m - 1)^{1/p}}. \quad (2)$$

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It is generated by an equal-amplitude, common-range, equiangular rank-one family. The cases $p = 1, 2, \infty$ are exact; in particular, $c_2(m) = \sqrt{(1 + \sqrt{m})/2}$.

At fixed matrix size, Zhang [3] introduced the notation $c_p^{\text{abs}}(m, n)$ and obtained the sharp endpoint values together with the general upper bound

$$c_p^{\text{abs}}(m, n) \leq \min\{m, n\}^{1/2-1/(2p)}.$$

Bourin and Lee [1] likewise highlighted the problem for Schatten exponents other than two. These results do not determine the rank-one optimum at $p = 3/2$.

For $n \geq 1$, define the rank-one restricted constant

$$c_{3/2}^{(1)}(n) = \sup_{\substack{A, B \in \mathbb{M}_n(\mathbb{C}), (A, B) \neq (0, 0) \\ \text{rank } A, \text{rank } B \leq 1}} \frac{\|A + B\|_{3/2}}{\| |A| + |B| \|_{3/2}}. \quad (3)$$

Our main result gives this constant exactly and shows that dimension two is already universal.

Theorem 1.1. *For every $n \geq 2$,*

$$c_{3/2}^{(1)}(n) = C_* := \left(\frac{107 + 51\sqrt{17}}{256} \right)^{1/6}. \quad (4)$$

The same value is obtained if the supremum is restricted to real matrices.

Put

$$s_* := \sqrt{\frac{\sqrt{17}-1}{8}}, \quad t_* := \sqrt{\frac{5\sqrt{17}-13}{8}}. \quad (5)$$

For nonzero A, B , equality holds in (4) if and only if, up to a common left unitary, a common right unitary, a common positive scale, interchange of A, B , and the harmless phase changes in rank-one factorizations, the pair is

$$A = e_1 e_1^*, \quad B = u_* y_*^*, \quad (6)$$

where, in a two-dimensional coordinate subspace,

$$y_* = \begin{pmatrix} s_* \\ \sqrt{1-s_*^2} \end{pmatrix}, \quad u_* = \begin{pmatrix} t_* \\ \sqrt{1-t_*^2} \end{pmatrix}.$$

Thus the problem in $\mathbb{M}_2(\mathbb{R})$ has a complete algebraic answer. The constant is the unique root in

$$\frac{2591}{2500} < C_* < \frac{2073}{2000} \quad (7)$$

of

$$128z^{12} - 107z^6 - 64 = 0. \quad (8)$$

This is the minimal polynomial of C_* over $\mathbb{Q}$: its primitive integer form is irreducible modulo 5, where its monic associate is $z^{12} + z^6 + 2$. The equality correlations are the unique positive roots in the isolating intervals

$$\frac{781}{1250} < s_* < \frac{6249}{10000}, \quad \frac{2439}{2500} < t_* < \frac{9757}{10000} \quad (9)$$

of $4s^4 + s^2 - 1$ and $4t^4 + 13t^2 - 16$, respectively.

The proof has two parts. First, a general factorization estimate removes the left rank-one vectors and the relative amplitude. Second, a scalar spectral quotient is maximized exactly. This also exhausts all degenerate faces of the original parameter domain; no interval subdivision is needed.

2. THE THREE-PARAMETER FORMULATION

Write two nonzero rank-one matrices as

$$A = a u x^*, \quad B = b v y^*, \quad (10)$$

where $a, b > 0$ and u, v, x, y are unit vectors. Common left and right unitaries, common scaling, interchange of A, B , and phase changes in the factorizations reduce the real problem to

$$r = \frac{b}{a} \in (0, 1], \quad s = |x^* y| \in [0, 1], \quad t = |u^* v| \in [0, 1],$$

together with the sign of $(u^* v)(x^* y)$. The larger sign is positive. Indeed, with $a = 1$, the squared singular values $\mu_{\pm}$ of $A + B$ have sum and product

$$T = 1 + r^2 + 2rst, \quad D = r^2(1 - s^2)(1 - t^2). \quad (11)$$

At fixed D , the function $\mu_+^{3/4} + \mu_-^{3/4}$ is strictly increasing in T . The two eigenvalues $\nu_{\pm}$ of $|A| + |B|$ satisfy

$$\nu_+ + \nu_- = 1 + r, \quad \nu_+ \nu_- = r(1 - s^2). \quad (12)$$

Consequently the direct objective is

$$\left(\frac{\|A + B\|_{3/2}}{\||A| + |B|\|_{3/2}} \right)^{3/2} = \frac{\mu_+^{3/4} + \mu_-^{3/4}}{\nu_+^{3/2} + \nu_-^{3/2}}. \quad (13)$$

The compactification $r \in [0, 1]$, $s, t \in [0, 1]$ contains every zero-amplitude and collinear boundary. Although (13) is algebraic after adjoining fourth roots, optimizing it directly obscures a stronger inequality that eliminates r and t simultaneously.

3. A FACTORIZATION BOUND

The next lemma is valid for arbitrary finite rank-one families and arbitrary dimension. We only need two summands.

Lemma 3.1 (Rank-one factorization bound). *Let $A_k = a_k u_k x_k^* \in \mathbb{M}_n(\mathbb{C})$, where $a_k \geq 0$ and u_k, x_k are unit vectors. Set*

$$M = \sum_k A_k, \quad H = \sum_k |A_k| = \sum_k a_k x_k x_k^*, \quad R = \text{Tr } H = \sum_k a_k.$$

Then

$$\|M\|_{3/2} \leq \sqrt{R \|H\|_3}. \quad (14)$$

Proof. By Schatten duality, for every Z with $\|Z\|_3 = 1$,

$$\begin{aligned} |\operatorname{Tr}(Z^* M)| &\leq \sum_k a_k |u_k^* Z x_k| \leq \sum_k a_k \|Z x_k\| \\ &\leq \left(\sum_k a_k \right)^{1/2} \left(\sum_k a_k \|Z x_k\|^2 \right)^{1/2} \\ &= \sqrt{R \operatorname{Tr}(Z^* Z H)}. \end{aligned}$$

Put $Q = Z^* Z$. Then

$$\|Q\|_{3/2} = (\operatorname{Tr} Q^{3/2})^{2/3} = \|Z\|_3^2 = 1.$$

Hölder's inequality for Schatten norms gives $\operatorname{Tr}(QH) \leq \|Q\|_{3/2} \|H\|_3 = \|H\|_3$. Taking the supremum over Z proves (14). $\square$

Remark 3.2. For every nonzero $H \geq 0$, equality in the last Hölder step forces

$$Q = \frac{H^2}{\|H\|_3^2}. \quad (15)$$

Indeed, this is the unique norming functional for nonzero H in the duality between the Schatten 3- and 3/2-classes; the statement also covers rank-deficient H . Equality in the weighted Cauchy–Schwarz step forces all nonzero quantities $\|Z x_k\|$ to coincide. These conditions will recover the amplitudes and both correlations.

4. THE SCALAR MAXIMUM

For two rank-one summands, H has at most two nonzero eigenvalues. Let $\lambda \geq \mu \geq 0$ denote them. Lemma 3.1 gives

$$\frac{\|A + B\|_{3/2}}{\|H\|_{3/2}} \leq \Psi(\lambda, \mu) := \frac{\sqrt{(\lambda + \mu)(\lambda^3 + \mu^3)^{1/3}}}{(\lambda^{3/2} + \mu^{3/2})^{2/3}}. \quad (16)$$

When $\mu = 0$, the right side equals one. For $\mu > 0$, write $\lambda = w^2 \mu$, $w \geq 1$. Homogeneity yields

$$\Psi(\lambda, \mu)^6 = h(w) := \frac{(w^2 + 1)^3 (w^6 + 1)}{(w^3 + 1)^4} = \frac{(w^2 + 1)^4 (w^4 - w^2 + 1)}{(w^3 + 1)^4}. \quad (17)$$

A direct differentiation factors completely:

$$h'(w) = -\frac{6w(w-1)(w^2+1)^3(w^4-w^3-2w^2-w+1)}{(w^3+1)^5}. \quad (18)$$

Put $z = w + w^{-1}$. For $w \geq 1$, $z \geq 2$ and is strictly increasing away from $w = 1$, while

$$\frac{w^4 - w^3 - 2w^2 - w + 1}{w^2} = z^2 - z - 4. \quad (19)$$

Thus h has a unique maximizer $w_* > 1$, characterized by

$$z_* = w_* + w_*^{-1} = \frac{1 + \sqrt{17}}{2}. \quad (20)$$

The sign in (18) shows that h increases before this point and decreases after it. In particular, this argument includes both endpoints $w = 1$ and $w = \infty$, where $h = 1$.

To evaluate the maximum without retaining w_* , observe that

$$h(w) = \frac{z^4(z^2 - 3)}{(z^3 - 3z + 2)^2}. \quad (21)$$

Using $z_*^2 = z_* + 4$ in (21) gives

$$h(w_*) = \frac{107 + 51\sqrt{17}}{256}. \quad (22)$$

Equations (16)–(22) prove the upper bound in Theorem 1.1.

5. SHARPNESS AND EQUALITY

We first construct an extremizer. Take unit vectors $x, y \in \mathbb{C}^2$ with $x^*y = s_*$, and set

$$H = xx^* + yy^*, \quad Z = \frac{H}{\|H\|_3}.$$

The eigenvalues of H are $1 + s_*$ and $1 - s_*$. From (20), the choice (5) is exactly the condition that their square-root ratio equals w_* . Also $\|Z\|_3 = 1$, and symmetry gives $\|Zx\| = \|Zy\|$. Define

$$u = \frac{Hx}{\|Hx\|}, \quad v = \frac{Hy}{\|Hy\|}. \quad (23)$$

For $A = ux^*$ and $B = vy^*$, every inequality in the proof of Lemma 3.1 is an equality. The dual pairing with Z therefore attains the upper bound, proving sharpness.

It remains to show that no equality cases were lost. Suppose $A = aux^*$ and $B = bvy^*$ attain C_* . The scalar maximum is strictly larger than one, so $a, b > 0$, the vectors x, y are independent, and H is positive definite on their span. Let Z be a norming matrix in Schatten duality. Equality in the proof of Lemma 3.1 and Remark 3.2 gives

$$Z^*Z = \frac{H^2}{\|H\|_3^2}, \quad \|Zx\| = \|Zy\|.$$

Writing $c = x^*y$, a direct computation gives

$$x^*H^2x - y^*H^2y = (a^2 - b^2)(1 - |c|^2). \quad (24)$$

Since $|c| < 1$, equation (24) forces $a = b$. The eigenvalues of H are then $a(1 + |c|)$ and $a(1 - |c|)$. The uniqueness of the scalar maximizer yields $|c| = s_*$.

By the polar decomposition of Z , equation (15) has the form

$$Z = W \frac{H}{\|H\|_3}$$

for a unitary W . Equality in the first two inequalities of Lemma 3.1 forces, after one common phase choice,

$$u = W \frac{Hx}{\|Hx\|}, \quad v = W \frac{Hy}{\|Hy\|}.$$

Changing the simultaneous phases of v, y if necessary, assume $x^*y = s_*$. Then

$$u^*v = \frac{s_*(3 + s_*^2)}{1 + 3s_*^2} = t_*. \quad (25)$$

Substitution of $s_*^2 = (\sqrt{17} - 1)/8$ in (25) gives $t_*^2 = (5\sqrt{17} - 13)/8$. Two ordered pairs of unit vectors with the same Gram matrix are related by a unitary on their spans. This proves the classification in (6) and completes the proof of Theorem 1.1.

Corollary 5.1 (Real 2×2 problem). *For real rank-at-most-one 2×2 matrices,*

$$\sup_{(A,B) \neq (0,0)} \frac{\|A + B\|_{3/2}}{\| |A| + |B| \|_{3/2}} = \left(\frac{107 + 51\sqrt{17}}{256} \right)^{1/6},$$

and all equality cases are the real instances of (6).

Remark 5.2 (Boundary exhaustion). The proof covers every face of the original compact three-parameter domain. If one amplitude vanishes, or if $s = 1$, then H has rank one and the scalar bound is $1 < C_*$. The cases $r = 1$, $s = 0$, and $t = 0, 1$ remain inside Lemma 3.1; equality there would have to satisfy the conditions just derived, which force $r = 1$, $s = s_*$, and $t = t_*$. Interchange of the summands handles $r = \infty$. Hence no separate limiting or singular case can attain the maximum.

6. STRICT COMPARISON WITH THE TANG–ZHANG VALUE

At $p = 3/2, m = 2$, let $x > 1$ solve $x^{3/2} - 2x - 1 = 0$, and put $\tau = \sqrt{x}$. Then

$$\tau^3 - 2\tau^2 - 1 = 0. \quad (26)$$

The Tang–Zhang value (2), denoted here by C_{TZ} , satisfies

$$C_{\text{TZ}}^6 = \frac{\tau^6}{16(\tau^2 + 1)}. \quad (27)$$

The following exact separation avoids decimal comparison.

Proposition 6.1. *One has*

$$C_{\text{TZ}} < \frac{207}{200} < C_*. \quad (28)$$

Numerically,

$$C_{\text{TZ}} = 1.0346539518514342\dots, \quad C_* = 1.0364143573334075\dots$$

Proof. The cubic in (26) is strictly increasing for $\tau > 4/3$, is negative at $4/3$, and

$$\left(\frac{1103}{500}\right)^3 - 2\left(\frac{1103}{500}\right)^2 - 1 = \frac{310727}{125000000} > 0.$$

Thus $\tau < 1103/500$. Since $u \mapsto u^6/[16(u^2 + 1)]$ is increasing for $u > 0$,

$$\left(\frac{207}{200}\right)^6 - \frac{(1103/500)^6}{16((1103/500)^2 + 1)} = \frac{26731151399029597}{18772595200000000000} > 0. \quad (29)$$

Equations (27) and (29) give the first inequality in (28).

For the second, put $q = 207/200$ and

$$D = 256q^6 - 107 = \frac{51922340886049}{250000000000} > 0.$$

The exact square difference

$$51^2 \cdot 17 - D^2 = \frac{67633016912924345597169599}{6250000000000000000000} > 0$$

shows $51\sqrt{17} > D$. Hence $(107 + 51\sqrt{17})/256 > q^6$, proving $C_* > q$. $\square$

7. ALGEBRAIC CERTIFICATE AND REPRODUCIBILITY

The exact constant and correlations have low-degree certificates:

$$\begin{aligned} 128C_*^{12} - 107C_*^6 - 64 &= 0, \\ 4s_*^4 + s_*^2 - 1 &= 0, \\ 4t_*^4 + 13t_*^2 - 16 &= 0. \end{aligned}$$

The rational intervals (7) and (9) isolate their positive roots. A supplementary standard-library Python script checks, using fractions only, the quotient-rule identity behind (18), all three algebraic relations and isolating signs, the irreducibility of (8) by Rabin's test over $\mathbb{F}_5$, the relation (25), and every rational margin in Proposition 6.1. A separate direct-SVD program reconstructs matrices, attacks both correlation signs and all boundary faces, and is used only as an adversarial check; no floating-point output enters the proof.

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